

RHINO SERFIC

FOUL WATER INSPECTION CHAMBER

SERFIC450 TO SERFIC1200 | DATA SHEET



OVERVIEW

The RHINO SERFIC series are prefabricated, factory-built foul water benched channelled inspection chambers, engineered for efficient foul water and drainage management. Each chamber has a main channelled base with optional side connections and integrates with the wider RHINO adaptor range.

Made from High-Density Polyethylene (HDPE) using extrusion and thermoforming, every chamber arrives fully assembled and ready to install, for light-duty through to full-traffic loading applications.

DESIGN COMPLIANCE

- BS EN 13598-2 (2009)
- Sewers for Adoption 7th Edition (SfA7)
- Design and Construction Guidance (DCG), including Parts B and C
- Building Regulations Part H1
- Adoptable and non-adoptable applications

BENCHED CHANNEL DESIGN

Sloped benching guides foul water, solids and sediment into the central flow channel rather than letting them settle in corners. The channel follows the profile of the connected pipes, so flow passes through with minimal turbulence and a lower risk of blockage, and the smooth benching is simple to clear during maintenance.

PIPEWORK	EN1401	TWINWALL
Main channel	110, 160	150, 225, 300
Side inlets	110, 160	150, 225, 300

Sizes in mm. Maximum inlet pipe size depends on chamber diameter - see page 3.

DIMENSIONAL DATA - SERFIC600 SHOWN

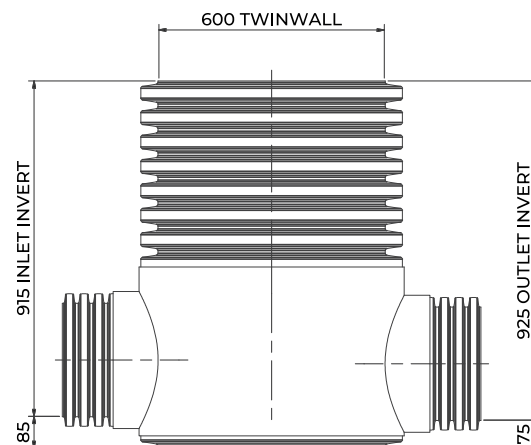
- Body: 600 mm twinwall
- Overall height: 1000 mm
- Top to inlet invert: 915 mm
- Top to outlet invert: 925 mm
- Inlet invert above base: 85 mm
- Outlet invert above base: 75 mm

CHAMBER OPTIONS

Six diameters from 450 to 1200 mm, each with a moulded benched base. Standard overall heights run from 1000 to 3000 mm in 500 mm steps, with inlet positions and pipe sizes configured per project. Adoptable installations up to 3000 mm to pipe soffit; non-adoptable up to 6000 mm.

INSTALLATION

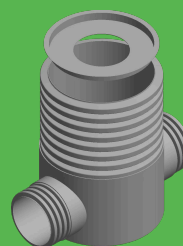
The installation design should consider traffic loadings, ground conditions and variable ground water levels. If in any doubt consult your scheme's designer or engineer. A suitably qualified professional is required to sign off the installation.



SERFIC600 elevation. Dimensions in mm. **NOT TO SCALE**

FEATURES & BENEFITS

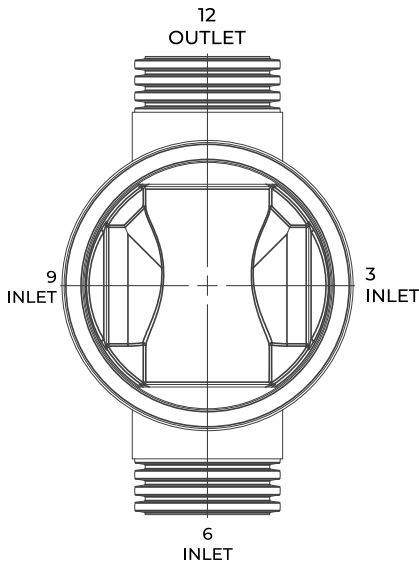
- One-piece HDPE unit, delivered fully assembled
- Up to 70% faster to install than sectioned chambers
- Up to five inlets at 3, 5, 6, 7 and 9 o'clock
- Outlet fixed at 12 o'clock
- Anti-fracture thermoformed base
- Resistant to hydrogen sulphide and corrosive acids and bases, critical in foul water service
- Light-duty to full-traffic loading
- Compatible with the full RHINO adaptor range



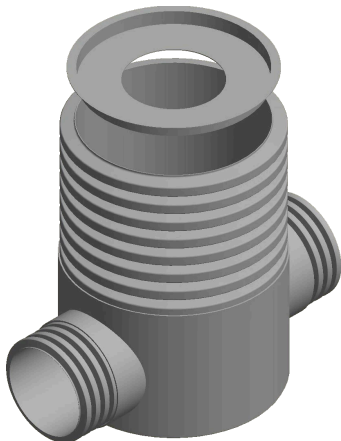
OPTIONAL REDUCTION CAP

A reduction cap is available at optional cost. It reduces the chamber opening to a Ø320 mm access, fitted to the top of the chamber shaft.

Use the online configurator at sudsenviro.com to specify diameter, inlets, pipe sizes and depth.



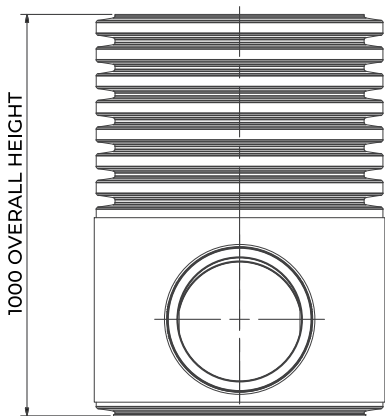
A PLAN VIEW NOT TO SCALE



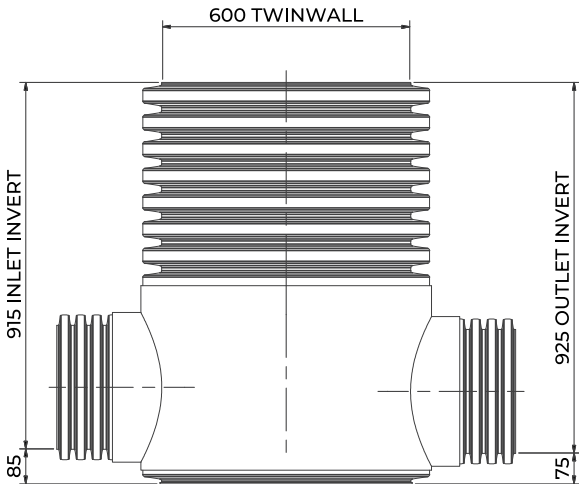
B 3D VIEW WITH OPTIONAL REDUCTION CAP (Ø320 OPENING)

NOTES

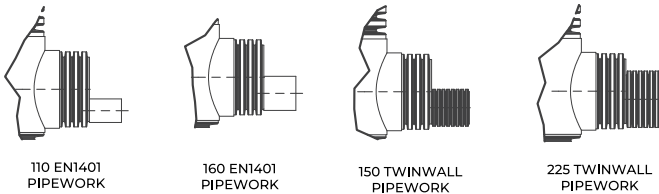
- 1. All dimensions in mm. Drawing not to scale; do not scale from this sheet.
- 2. Outlet fixed at 12 o'clock. Inlets shown at 3, 6 and 9 o'clock.
- 3. Inlet invert 85 mm and outlet invert 75 mm above the chamber base, giving a 10 mm fall across the chamber.
- 4. Chamber body 600 mm twinwall, overall height 1000 mm as drawn. Other heights to order.
- 5. Side connections take the adaptors shown in view E. Other pipework systems available.
- 6. Reduction cap supplied at optional cost.
- 7. Third angle projection.



C SIDE ELEVATION D SECTION ELEVATION THROUGH INLETS NOT TO SCALE



PIPEWORK OPTIONS
OTHER PIPEWORK SYSTEMS AVAILABLE



E SIDE CONNECTION PIPEWORK OPTIONS NOT TO SCALE



RHINO SERFIC
INSPECTION CHAMBER

PRODUCT CODE SERFIC600	DRAWING REF SERFIC600300-1
DESCRIPTION 600 mm, 3 inlets, 1000 mm	SCALE Not to scale
UNITS Millimetres	PROJECTION Third angle

SPECIFICATION BY DIAMETER

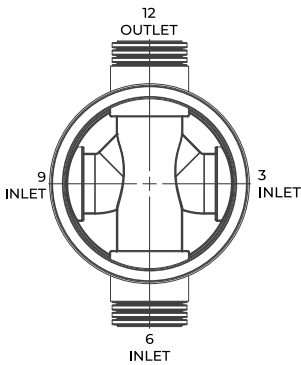
PRODUCT CODE	CHAMBER Ø	MAX INLETS	MAX INLET PIPE	ADOPTABLE DEPTH	NON-ADOPTABLE DEPTH
SERFIC450	450 mm	2	160 mm EN1401	3000 mm	6000 mm
SERFIC600	600 mm	4	225 mm twinwall	3000 mm	6000 mm
SERFIC750	750 mm	5	300 mm twinwall	3000 mm	6000 mm
SERFIC900	900 mm	5	300 mm twinwall	3000 mm	6000 mm
SERFIC1050	1050 mm	5	300 mm twinwall	3000 mm	6000 mm
SERFIC1200	1200 mm	5	300 mm twinwall	3000 mm	6000 mm

Depths to pipe soffit (adoptable: DCG / SfA7). Inlets at 3, 5, 6, 7 and 9 o'clock; outlet at 12 o'clock. With three or more inlets on a 600 mm chamber, or four or more on any chamber, the outlet is a minimum of 225 mm twinwall. Inlets are never larger than the outlet. Drawing references extend the code with pipework size and overall height in metres, for example SERFIC600300-1.

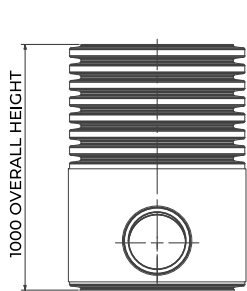
DRAWN SIZES

The 600 mm chamber is drawn in two pipework variants, SERFIC600225 (overall heights 0.5 to 4 m) and SERFIC600300 (0.5 to 3 m). At 1 m both show 915 mm and 925 mm inverts, 85 mm and 75 mm above base.

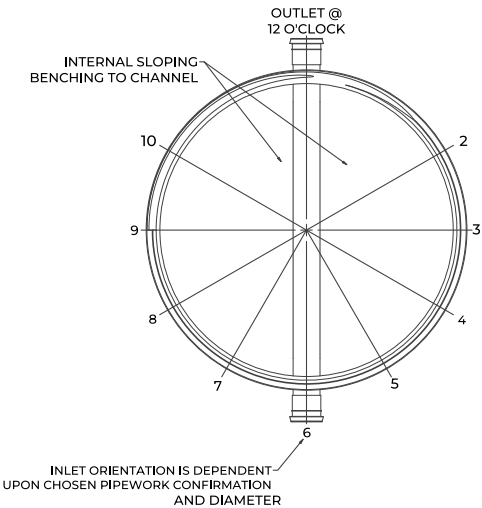
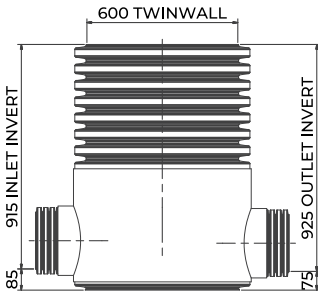
Drawings for the 750 to 1200 mm chambers carry the FIC reference, for example FIC1200110-600.



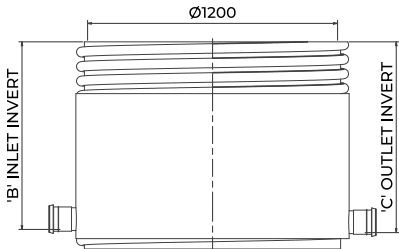
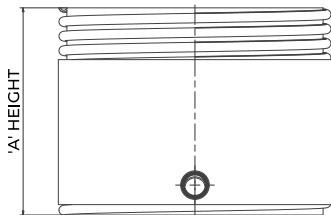
F SERFIC600225-1.0 PLAN
NOT TO SCALE



G SERFIC600225-1.0 ELEVATIONS, 1000 MM HIGH
NOT TO SCALE



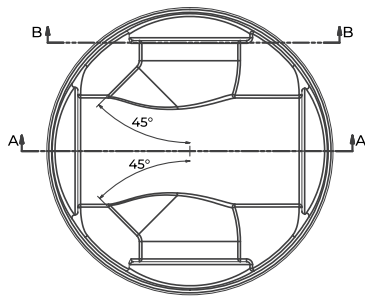
H 1200 MM PLAN AND CLOCK POSITIONS
NOT TO SCALE



J 750 TO 1200 MM RANGE: A HEIGHT, B INLET INVERT, C OUTLET INVERT
NOT TO SCALE

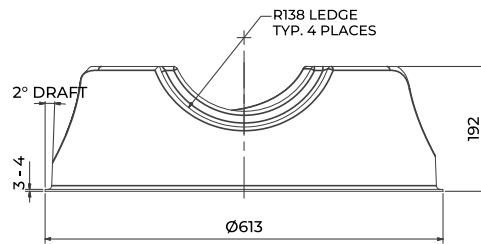
A HEIGHT	B INLET INVERT	C OUTLET INVERT	FALL
1000	895	910	15
1500	1395	1410	15
2000	1895	1910	15
2500	2395	2410*	15
3000	2895	2910	15

mm, measured from the top of the chamber. Applies to 750, 900, 1050 and 1200 mm chambers.
*Source range table reads 1510; 2410 follows the 90 mm offset of every other row. To be confirmed.



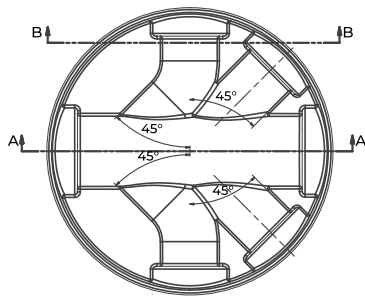
K 3-INLET BASE SEB600-225-225

NOT TO SCALE



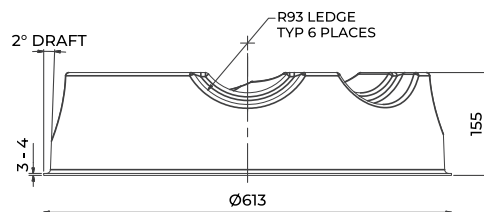
ELEVATION: Ø613 X 192 HIGH, R114 CHANNELS

NOT TO SCALE



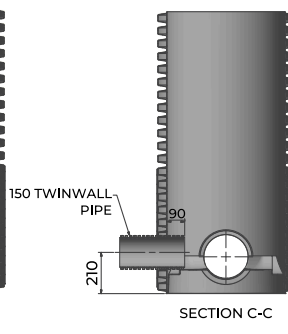
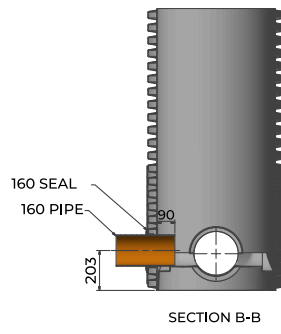
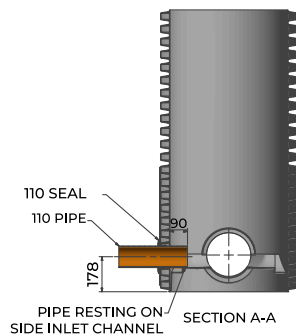
L 5-INLET BASE SEB600-150-150-150

NOT TO SCALE



ELEVATION: Ø613 X 155 HIGH, R75 CHANNELS

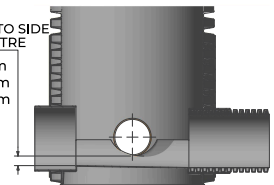
NOT TO SCALE



M SIDE INLET SECTIONS, 600 MM CHAMBER

NOT TO SCALE

MAIN INLET
40mm DIFFERENCE TO SIDE
INLET DRILLING CENTRE
110 Pipe = 138mm
160 Pipe = 163mm
150 Pipe = 170mm



SECTION D-D

N MAIN INLET SECTION

NOT TO SCALE

INSTALLATION

- Delivered as one piece, ready for connection and backfill.
- Design for traffic loading, ground conditions and variable ground water levels.
- Maintain a minimum 50 mm fall from side inlets to the main channel (DCG Parts B and C).
- Where depth exceeds 1 m, fit a restricted access cover (DCG).
- A suitably qualified professional must sign off the installation.

COMPLIANCE

BS EN 13598-2	Thermoplastic inspection chambers (2009)
SfA 7th Edition	Adoptable sewer applications
DCG	Adoptable and non-adoptable systems
Part H1	Building Regulations, drainage